

Application of Derivatives

Advanced Level
Combined Mathematics
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PART - 1

Tangent line and Normal line drawn to a curve.

❖ Find the equations of the tangent and the normal to the following curves at the indicated points :

01. $y = 2x^3 + x^2 - 7 - 1$ at $P(2, 5)$

02. $y = x^2 + 5x^3 + 8x^2 + 7x + 3$ at $P(-2, -3)$

03. $y^2 = \frac{x^2}{(5-x)}$ at $P(4, -4)$

04. $y^2 = 4ax$ at $P(at^2, 2at)$

05. $ay^2 = x^2$ at $P(am^2, am^3)$

06. $y = 3 \sin x + 5 \cos x$ at $x = 0$

07. $y = \cot^2 x - 2 \cot x + 2$ at $x = \frac{\pi}{4}$

08. $x = (1 - \cos \theta), y = (\theta - \sin \theta)$ at $\theta = (\pi/2)$

09. $x = t^2$ and $y = t^3$ at $t = 2$.

10. Find the equations of the tangent and the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$.

11. Find the equations of the tangent and the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$.

12. Find the equations of the tangent and the normal to the parabola $y^2 = 4ax$ at the point $P(x_1, y_1)$.

13. Find the equations of the tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point x_1, y_1 and show that the sum of its intercepts on the co-ordinate axes is constant.

14. Find the equations of normal to the curve $y = 2 \sin^2 3x$ at $x = (\pi/6)$
15. Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel to $x - \text{axis}$.
16. Find the equation of the tangent on the curve $y^2 = 4x + 5$, which is parallel to the line $y = 2x + 1$.
17. Find the equation of the tangent on the curve $y = \sqrt{3x - 2}$ which is parallel to the line $4x - y + 5 = 0$
18. Find the equation of the normal to the curve $3x^2 - y^2 = 8$, which is parallel to the line $x + 3y = 5$.
19. Find the point on the curve $y = 2x^2 + 3x + 5$, the tangent at which is perpendicular to the line $5y - x = 10$.
20. Find the slope of the tangent to the curve $x = t^2 + 3t - 8, y = 2t^2 - 2t - 5$ at the point $(2, -1)$ and hence find the equation of the tangent at this point.
21. Find the point on the curve $y^2 = x$, the tangent at which makes an angle of 45° with the $x - \text{axis}$.
22. Find the point on the curve $y = x^2 + 3x + 4$, the tangent at which pass through the origin.
23. Show that the line $\frac{x}{a} + \frac{y}{b} = 1$ touches the curve $y = be^{-x/a}$ at the point where the curve crosses the axis of y .
24. Show that the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ touches the line $\frac{x}{a} + \frac{y}{b} = 2$ at the point (a, b) for all values of n .
25. If the straight line $x \cos \alpha + y \sin \alpha = p$ touches the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that $a^2 \cos^2 \alpha + b^2 \sin^2 \alpha = p^2$.
26. If $x \cos \alpha + y \sin \alpha = p$ touches the curve $x^m y^n = a^{m+n}$,
Prove that $p^{m+n} m^m n^n = (m + n)^{m+n} a^{m+n} \cos^m \alpha \sin^n \alpha$.
27. Prove that the tangent to the curve $y = x^2 - 5x + 6$ at the points $A(2, 0)$ and $B(3, 0)$ are at right angles.
28. Find that point the parabola $y = (x - 3)^2$, where the tangent is parallel to the line joining the points $A(5, 4)$ and $B(3, 0)$.
29. Find that point on the curve $y = 2x^2 - 6x + 9$, at which the tangent is parallel to $x - \text{axis}$.

30. Show that the curves $y^2 = 2x$ and $2xy = k$ cut at right angles, if $k^2 = 8$.
31. Show that the curves $x = y^2$ and $xy = k$ cut at right angles, if $8k^2 = 1$.
32. Find the angles of intersection of the curves $y^2 = x$ and $x^2 = y$.
33. Find the angles of intersection of the curves $y = 4 - x^2$ and $y = x^2$.
34. Show that the curves $xy = a^2$ and $x^2 + y^2 = 2a^2$ touch each other.
35. Show that the curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - 2 = 0$ cut orthogonally.

Past Paper Questions

36. Show that the equation of the tangent and the normal to the equation $y = 2x^2 + 3x - 5$ at $x = 1$ is given by $y - 7x + 7 = 0$ and $7y + x - 1 = 0$ respectively.
37. Find the pair of equations of the tangents and the normal, drawn to the curve $y^2(2 - x) = x^3$ at $(1, 1)$ and $(1, -1)$.
38. Find the equation of the tangent and the normal drawn to the curve parametrically given by $x = \frac{1}{t}, y = t - \frac{1}{t}$ at $t = 1$.
39. Let p and q be real constants. The equation of the tangent drawn to the curve, $y = p + 2qx + 2x^3$ at the point where the curve meets the Oy axis is, $2x - 3y + 1 = 0$. Show that, $p = q = \frac{1}{3}$.
40. Value of the gradient of the tangent drawn to the curve $y = 2px^2 + \frac{3q}{x}$ at $(1, 2)$ is 3. Show that the values of the real constants $p = \frac{5}{6}$ and $q = \frac{1}{9}$.
41. Two curves $y = e^{-x}$ and $y = e^{kx}$ are intersect at right-angles. Find the value of $k \in \mathbb{R}$.
42. Let t be a parameter and p be a real constant. A curve is defined parametrically by $x = p(\cos t + t \sin t), y = p(\sin t - t \cos t)$. Show that the perpendicular distance between the origin O and normal drawn to the curve at any point is a constant.
43. If the curves $ax^2 + by^2 - 1 = 0$ and $px^2 + qy^2 - 1 = 0$ are orthogonal to each other for all x and y values, show that $pq(a - b) = ab(p - q)$.

44. A curve is given by $x = 3t, y = \frac{3}{t}$, where t is non zero parameter. Show that the equation of the tangent drawn to the curve at $\left(3t, \frac{3}{t}\right)$ is $x + t^2y = 6t$. Deduce that as t varies, the triangular area enclosed by the curve and the co-ordinate axes remains a constant.
45. A curve C is given by the equation $y = 4 - 4x + 3x^2 - x^3$. Find the equation of the tangent drawn to the curve C at the point $(1, 2)$. Show that the tangent is perpendicular to the tangent drawn to the curve $y^2 = 4x$ at the point $(1, 2)$.
46. Let C be the curve given by $x = 2 \cos \theta$ and $y = \sin \theta$, where θ is a parameter. The normal to the curve C at the point corresponding to $\theta = \frac{\pi}{4}$ meets the curve C at the point corresponding to $\theta = \alpha$. Show that, $2 \sin \alpha - 8 \cos \alpha + 3\sqrt{2} = 0$.
47. Let C be the curve given by $x = e^t + e^{-t}, y = e^t - e^{-t}$, where t is a real parameter. Find $\frac{dy}{dx}$ in terms of t and show that the equation of the tangent line at the point on C corresponding to $t = \ln 2$ is $5x - 3y - 8 = 0$.
48. A curve C in the xy -plane is given, in terms of a real parameter θ , by the equation $x = 2 + \cos 2\theta, y = 4 \sin \theta$. Find the derivative $\frac{dy}{dx}$ in terms of θ , and show that the equation of the normal drawn to the curve C at the point where, $\theta = \frac{\pi}{4}$ is $x - \sqrt{2}y + 2 = 0$.
49. A curve C is given by the parametric equations $x = 3 \sin^3 \left(\frac{\theta}{2}\right), y = \sin^3 \theta$ for $0 < \theta < \frac{\pi}{4}$. Show that, $\frac{dy}{dx} = \sin 2\theta$. If the gradient of the tangent at a point P on C is $\frac{\sqrt{3}}{2}$, find the value of the perimeter θ corresponding to P .
50. A curve C is given parametrically by $x = 3 \cos \theta - \cos^3 \theta, y = 3 \sin \theta - \sin^3 \theta$, for $0 < \theta < \frac{\pi}{2}$. Show that, $\frac{dy}{dx} = -\cot^3 \theta$. Find the co-ordinates of the point p on the curve C at which the gradient of the tangent line is -1 .
51. A curve C is given parametrically by $x = 3 \cos \theta - \cos^3 \theta, y = 3 \sin \theta - \sin^3 \theta$, for $0 < \theta < \frac{\pi}{2}$. Show that, $\frac{dy}{dx} = -\cot^3 \theta$. Find the co-ordinates of the point p on the curve C at which the gradient of the tangent line is -1 .
52. A curve C is given parametric equations $x = \ln \left(\tan \frac{t}{2}\right)$ and $y = \sin t$ for $\frac{\pi}{2} < t < \pi$. Show that, $\frac{dy}{dx} = \cos t \sin t$. Deduce that, the gradient of the tangent line drawn to the curve C at the point corresponding to $t = \frac{2\pi}{3}$ is $-\frac{\sqrt{3}}{4}$.

53. Let C be the curve given parametrically by $x = \ln t$ and $y = e^3 + t \ln t$ for $t > 0$.
 Show that, $\frac{dy}{dx} = t(e^3 + \ln t + 1)$.
 If the tangent drawn to the curve C at the point corresponding to $t = 1$, passes through the point $(1, a)$, show that $a = 1 + 2e$.
53. Let C be the parabola parametrically given by $x = at^2$ and $y = 2at$ for $t \in \mathbb{R}$, where $a \neq 0$.
 Show that the equation of the normal line to the parabola C at the point $(at^2, 2at)$ is given by $(y + tx = 2at + at^3)$.
 The normal line at the point $P \equiv (4a, 4a)$ on the parabola C meets this parabola again at a point $Q \equiv (aT^2, 2aT)$. Show that $T = -3$.
54. Show that equation of the tangent line to the rectangular hyperbola parametrically given by $x = ct$ and $y = \frac{c}{t}$ for $t \neq 0$ at the point $P \equiv (cp, \frac{c}{p})$ is given by $x + p^2y = 2cp$.
 The normal line to this hyperbola at P meets the hyperbola again at another point $Q \equiv (cp, \frac{c}{p})$.
 Show that, $p^3q = -1$.
55. The tangent to the curve $v(1 + x^2) = 2$ at the point $P(3, \frac{1}{5})$ meets the curve again at Q .
 Find coordinates of Q .
56. The tangent at the point $P(at^2, at^3)$ to the curve $ay^2 = x^3$, where a is a constant meets the curve again at Q . Find the coordinates of Q in terms of t .
57. Let C be the curve given parametrically by $x = \frac{a}{2}(t + \frac{1}{t})$ and $y = a(t - \frac{1}{t})$, where a is a non-zero constant and t is a non-zero parameter. Find the equation of the normal to the curve C at the point with parametric value t_0 .
 Show that four normals can be drawn from the point $(-13a, 0)$, to the curve C and find the parametric values of the feet of the four normals.

PART - 2

Sketch the graph of following functions.

58. $y = \frac{x^3}{3} - \frac{x^2}{2}$

59. $y = \frac{x^3}{3} - x^2 - 3x$

60. $\frac{2x^3}{3} - \frac{3x^2}{2} - x$

61. $\frac{x^4}{4} - \frac{x^3}{3}$

62. $\frac{(x-1)^4}{4} + \frac{(x-1)^3}{3}$

63. If $f(x) = x^2 - 2x - 5$, find the range of x in which the function f increasing and decreasing. Hence sketch the graph of $y = f(x)$.

64. If, $f(x) = x^3 - 12x + 3$, find the range of x in which the function f increasing and decreasing. Hence sketch the graph of, $y = f(x)$.

65. Let $f(x) = 3x^5 - 5x^3$. Show that, f has three stationery points. Find the behavior of $f(x)$ at these points.

66. Let $f(x) = 3x^2 - 4x$. Sketch the graph of $y = f(x)$.

67. Show that function $f(x) = \frac{2x-1}{(x+1)^2}$ has vertical and horizontal asymptotes.

Show with usual notations, that $f'(x) = \frac{-2(x-2)}{(x+1)^3}$ and assume that, $f''(x) = -2 \frac{(-2x+72)}{(x+1)^4}$.

Find the maximum and minimum value of $f(x)$. Identify all the points of inflection.

Sketch the graph of $y = f(x)$.

68. Sketch the graph of $(x) = \frac{x^2-3x}{x-1}$

Hence sketch the graph of $f = \frac{1}{f(x)}$

69. Show that function $f(x) = \frac{x^2-x+1}{x-1}$ has vertical and horizontal asymptotes.

Show with usual notations, that $f'(x) = \frac{x^2-2x}{(x-1)^2}$ and assume that, $f''(x) = 2 \frac{1}{(x-1)^3}$.

Find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$.

70. Let $f(x) = \frac{x^2+2}{x^2-4}$. Show with usual notations, that $f'(x) = -\frac{12x}{(x^2-4)^2}$ and assume that, $f''(x) = 12 \frac{(-3x^2-2)}{(x^2-4)^3}$.

Find the maximum and minimum values of $f(x)$ using $f'(x)$ identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$ and hence $y = |f(x)|$ and $y = -f(x)$.

71. Let $f(x) = \frac{3-x}{(x+1)(x-2)}$; $x \neq -1, 2$. Show that, $f'(x)$ the derivative of $f(x)$ is given by $f'(x) = \frac{(x-5)(x-1)}{(x+1)^2(x-2)^2}$; $x \neq -1, 2$. Sketch the graph of $y = f(x)$ indicating the asymptotes and the turning points. Using the graph find the values of k such that, $k(x+1)(x-2) = 3-x$ has no real roots.

72. Let $f(x) = \frac{x^2-4}{x-1}$. Show with usual notations, that $f'(x) = \frac{x^2-2x+4}{(x-1)^2}$ and assume that, $f''(x) = -6 \frac{1}{(x-1)^3}$. Find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$ sketch the graph of $y = f(x)$.

73. Let $f(x) = \frac{x}{x^2-1}$; $x \neq 1, -1$. Show that $f'(x) = \frac{-(x^2+1)}{(x^2-1)^2}$; $x \neq 1, -1$. Sketch the graph of $y = f(x)$. Assuming $f''(x) = \frac{2x^3+6x}{(x^2-1)^3}$, find all the points of inflection of $f(x)$.

74. If, $f(x) = \frac{x^3(x-1)^2}{(2x+1)^2}$, show that, with usual notations, $f'(x) = \frac{3x^2(2x-1)(x+1)(x-1)}{(2x+1)^3}$; $x \neq -\frac{1}{2}$ and assume that, $f''(x) = \frac{6x(4x^2+4x^3-2x^2-4x+1)}{(2x+1)^4}$; $x \neq -\frac{1}{2}$.

Find the maximum and minimum values of $f(x)$ using $f'(x)$.

Sketch the graph of $y = f(x)$. Identify all the points of inflection using $f''(x)$.

75. If, $f(x) = \frac{x^2+1}{x-1}$, find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$.
76. Let $f(x) = \frac{x}{2} - \sin x$.
Find the region in which $y = f(x)$ increasing and decreasing in $0 = 2\pi$.
Sketch the graph of $y = f(x)$.
77. Let, $f(x) = \frac{x^2}{x-1}$. Show with usual notations, that $f'(x) = \frac{x^2-2x}{(x-1)^2}$ and assume that, $f''(x) = 2 \frac{1}{(x-1)^3}$. Find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$. Hence find the values of $k(\in \mathbb{R})$ for which the equation $x^2 - kx + k = 0$ has no real solutions.
78. If, $f(x) = \frac{x}{x-1}$, show with usual notations, that $f'(x) = -\frac{1}{(x-1)^2}$. Assume that, $f''(x) = 2 \frac{1}{(x-1)^3}$. Find the maximum / minimum points and points of inflection of $f(x)$.
Sketch the graph of $y = f(x)$.
79. Let $f(x) = \frac{x^2-2x+4}{x-2}; x \neq 2$. Show that with usual notations, $f'(x) = \frac{x(x-4)}{(x-2)^2}; x \neq 2$ and $f''(x) = \frac{8}{(x-2)^3}; x \neq 2$. Show that f has a local maximum at 0 and local minimum at 4. Identify all the points of inflection using $f''(x)$. Sketch proudly and graph of $y = f(x)$.
80. If, $(x) = x - \frac{1}{x}$, Show that with usual notations, $f'(x) = 1 + \frac{1}{x^2}; x \neq 0$ and $f''(x) = -\frac{2}{x^3}; x \neq 0$. Hence find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$.
81. If, $(x) = \frac{x^2+4}{3x^2}$, show that with usual notations, $f'(x) = \frac{2(x^4-1)}{3x^3}; x \neq 0$ and $f''(x) = \frac{2(x^4+3)}{3x^4}; x \neq 0$. Find the maximum and minimum values of $f(x)$ using $f'(x)$. Identify all the points of inflection using $f''(x)$. Sketch the graph of $y = f(x)$.
82. Using the first derivative, find the maximum / minimum points and points of inflection of $y = x^3 + 1$. Sketch the graph of $y = x^3 + 1$.
83. Use the derivative identify maximum, minimum points and the points of inflection of the curve $y = x^2 - 4x^3$. Sketch the graph of y .

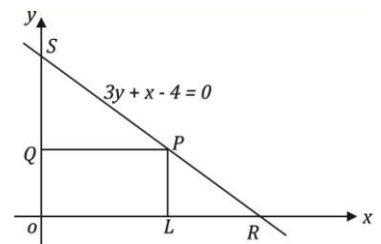
84. Identify maximum, minimum points and the points of inflection of $y = x^3 - 3x^2 - 9x + 15$. Sketch the graph of y .
85. Find the maximum and minimum points of $f(x) = 5x^3 - 3x^5$. Using the first derivative of $f(x)$. Use $f''(x)$ to identify all the points of infection. Sketch the graph of $f(x) = 5x^3 - 3x^5$. Using the graph,
- sketch $y = |f(x)|$.
 - sketch $y = -f(x)$
- Find the value of $k(\in \mathbb{R})$ for which the equation $5x^3 - 3x^5 - k = 0$ has only one real solutions.
86. Sketch Graph of, $f(x) = \frac{1}{3}x^3 - x^2$.
87. Sketch the graph of following functions.
- Let $f(x) = \frac{x^2}{x^2+1}$. Show with usual notations that the first derivative, $f'(x) = \frac{2x}{(x^2+1)^2}$. Using the $f'(x)$, find the maximum and minimum points of $f(x)$. Use $f''(x)$ to identify all the point of inflection. Sketch the graph of $y = f(x)$.
 - Let $f(x) = \frac{x^2}{x^2+4}$. Show that with usual notations, $f'(x) = \frac{8x}{(x^2+4)^2}$ and $f''(x) = -8 \frac{(3x^2-4)}{(x^2+4)^3}$. Using the $f'(x)$, show that f has a local minimum at $x = 0$. Use $f''(x)$ to identify all the points of inflection. Sketch roughly the graph of $y = f(x)$.
 - If, $f(x) = \frac{x}{9x^2+1}$ sketch the graph of $y = f(x)$. Show that with usual notations, $f'(x) = \frac{-9x^2+1}{(9x^2+1)^2}$ and assume that, $f''(x) = -54x \frac{(-3x^2+1)}{(9x^2+1)^3}$. Identify all the point of inflection. Using the graph,
 - Sketch $y = |f(x)|$.
 - Sketch $y = -f(x)$
 - sketch $= \frac{1}{f(x)} = \frac{9x^2+1}{x}$
 - find the value of $k(\in \mathbb{R})$ for which the equation $9kx^2 - x + k = 0$ has no real solutions.

PART - 3

Maximum and Minimum.

88. Using the differentiation discuss the behavior of $\left(\frac{x}{\sin x}\right)$ for all x values of $0 < x < \frac{\pi}{2}$.
89. Discuss the regions in which the function $f(x) = 2x - \tan x$ increasing and decreasing within the region $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Sketch the graph of $y = f(x)$.
90. Discuss the regions in which the function, $f(x) = 2 \cos x - \cos 2x$ increasing and decreasing within the region $0 < x < 2\pi$. Sketch the graph of $y = f(x)$.
91. Find the dimensions of the rectangle of largest area that has its base on the x -axis and its other two vertices above the x -axis and laying on the parabola $y = 12 - x^2$.

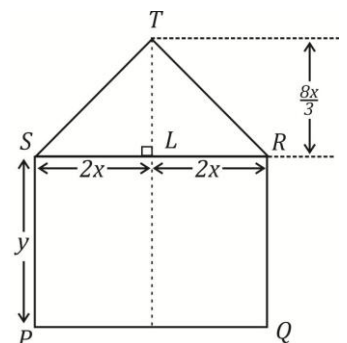
92. Point P is vary on the line $3y + x - 4 = 0$ between SR , as shown by the diagram. Find the dimension of the rectangle $PQOL$ when it's area becomes maximum (draw)



93. Find the dimension of the rectangle which can inscribe inside the circle bounded by, $x^2 + y^2 = 7$ and the Ox -axis.
94. Three town A, B and C are located at the vertices of an isosceles triangle such that the distance from A to B and A to C are equal. The distance from B to C is 12 km and the altitude through A is 16 km. How far from A , along the altitude through A , should a well be located so that it will use minimum pipe when supplying water to all three towns A, B and C .

95. A thin sheet of metal is cut as shown by the diagram, with $PQRS$ is a rectangular sheet and TSR is a triangular sheet with $TR = TS$. If, the perimeter of the adjacent diagram is 60 cm, by taking y cm, $RS = 4x$ cm and $TL = \frac{8x}{3}$ cm, find,

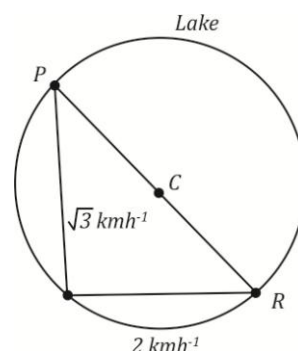
- i. An expression for perimeter l and area A in terms of if x and y .
- ii. The maximum value of A and the corresponding value of x .



96. Three points A, B and C are located on a circumference of a circle whose radius is r such that, $AB = AC$. If $\alpha = \hat{BAC}$ write down an expression for the area S of the triangle ABC in terms of r and α . Show that when S is maximum $\alpha = \frac{\pi}{3}$.

97. Saman is off-shore in a boat and wishes of reach a coastal village which is 6 km down a straight shoreline from the point on the shore to the nearest to the boat. He can row at 2 kmh^{-1} and run at 5 kmh^{-1} . Where should he land his boat to reach the village in the least amount of time?

98. C is the centre of a circle circular lake whose radius is $\sqrt{3}$ km. A boy start to swim a straight course with uniform velocity $\sqrt{3} \text{ kmh}^{-1}$ from point P to Q as shown by the diagram. Then he walks the arc QR with uniform velocity 2 kmh^{-1} where PR is a diameter of the lake. If $\angle QPR = \theta$ find the total time taken by the boy to reach R from P , in terms of θ . Hence find the minimum value of T .



99. Three towns A, B and C are connected by two straight road AB and BC such that, $\angle ABC = \frac{\pi}{2}$, $AB = 15 \text{ km}$ and $BC = 50 \text{ km}$. There is a proposed project to build another straight rod connecting the town A to place D on the road BC . Maximum speeds allowed for a car on the section DC is 50 kmh^{-1} and on the proposed road AD is 40 kmh^{-1} . If D is located $x \text{ km}$ from the town A , find the $T(x)$ the total time taken in hours by a car to travel A to C via D . Assuming that it travels at the maximum permitted speeds examine the sign of $\frac{dT}{dx}$ as x increases form 0 to 50 km. Find the most suitable place for D that will allow a car to complete the journey from A to C in the shortest time.

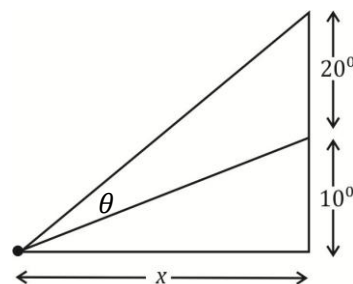
100. Find the dimensions of (radius r , height h)
- The right circular cylinder
 - The right circular cone, of maximum volume which can be inscribed in a radius a .
101. Find the dimension of the largest right angled triangle with a the given hypotenuse.
102. A circular cone with slanted height l (constant) and semi vertical angle α is to be made. Show that, $\tan \alpha = \sqrt{2}$ when the volume of the cone gets maximum.
103. A thin wire of length l is bent to form the complete perimeter of a sector of a circle of radius r .
- If the area of the sector is A , find A in terms of r .
 - Hence find the radius of the sector when its area is a maximum, showing that this area is the maximum.
 - Show that the respective angle is 2 radians.

104. A wire of given length l is cut in to two portions. One portion is bent in to the shape of a circle and the other portion in to the shape of a square. Show that, $A(x)$, the sum of the areas of the circle and the square is given by $A(x) = \frac{x^2}{4\pi} + \frac{(l-x)^2}{16}$ square units, where x , $(0 \leq x \leq l)$ is the length of the portion of the wire that is bent in to the form of the circle. Hence, show that area $A(x)$ is maximum when the side of the square is equal to the diameter of the circle.

105. Summation of two positive integers is 20. Find the two number such that the product of, square of one number and cube of other number gets maximum.

106. A movie screen on a wall is 20 feet height and 10 feet above the floor. At what distance x from the front of the room should you position yourself, so that the viewing angle θ of the movie screen is as large as possible.

Find the maximum value of θ .

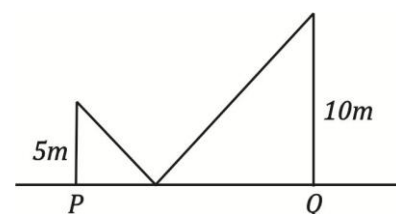


107. An open box (without lid) with square base is formed with thin aluminum sheet with area 192 cm^2 . Find the dimension of the box if its volume is maximum. In that case deduce that, the volume of the box is 192 cm^3 .

108. Let the summation of two integers be 20. Find the two integers so that, multiplication of two numbers is highest as possible.

109. Find the co-ordinate of point on the curve $y = -x^2 + 9$ to the point $(0, 3)$ so that the distance between two points is minimum. Find also the shortest distance.

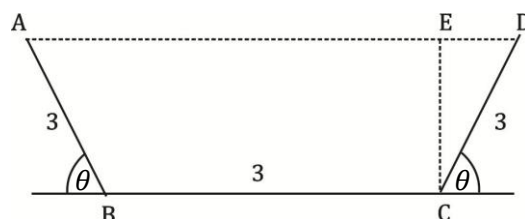
110. Two communication towers P and Q of height 5 m are to be located vertically 10 m apart on a horizontal ground. To fix the towers vertically a rope is attached to the top of the tower via a peg fixed between the towers on the ground. Show that, length of the rope used is minimum if the location of the peg is $\frac{10}{3} \text{ m}$ from the tower P .



111. A straight piece of wire of 28 cm is cut into two pieces. One piece is bent into a square (i.e. dimensions x times x .) The other piece is bent into a rectangle with aspect ratio three (i.e. dimensions y times $3y$.) What are dimensions, in centimeters, of the square and the rectangle such that the sum of their areas is minimized?
112. A straight piece of wire 40 cm long is cut into two pieces. One piece is bent in-to a circle and the other is bent in-to a square. How should wire be cut so that the total area of both circle and square is minimized?
113. A rectangle with sides parallel to the co-ordinate axes is to be inscribed in the region enclosed by the graphs of $y = x^2$ and $y = 4$ so that its perimeter has maximum length.
- Sketch the region under consideration.
 - Supposing that the x co-ordinates of the bottom right vertex of the rectangle is a find a formula which expresses P , the length of the perimeter, in terms of a .
 - Find the value of a which gives the maximum value of P , and explain why you know that this value of a gives a maximum.
 - What is the maximum value of P , the length of the perimeter of the rectangle?
114. The top and bottom margins of the poster are each 3 cm, and the side margins are each 2 cm. If the area of the printed material on the poster (that is, the area between the margins) is fixed at 384 cm^4 , find the dimensions of the poster with the smallest total area. Deduce that the minimum area is 600 cm^2 .
115. A hollow plastic cylinder with a circular base and open top is to be made and 10 m^2 plastic is available. Find the dimensions of the cylinder that give the maximum volume and find the value of the maximum volume.
116. An open-top box is to have a square base and a volume of 10 m^3 . The cost per square meter of material is 5 rupees for the bottom and 2 rupees for the four sides. Let x and y be lengths of the box's width and height respectively. Let C be the total cost of material required to make the box.
- Express C as a function of x and find its domain.
 - Find the dimensions of the box so that the cost of materials is minimized.
 - What is this minimum cost?

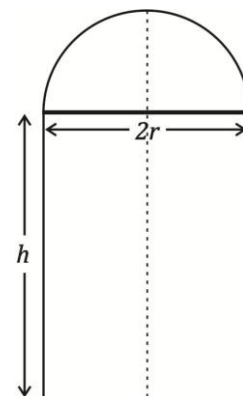
117. An open rectangular box is made from a square sheet of cardboard of side $2l$ cm by removing a square of side x cm from each of the corners and joining the cut edges. If the volume of the box is v cm³ find,
- An expression for v in term of l and x .
 - The maximum value of v in terms of l , justifying that the value of v is a maximum.
118. What angle θ between two edge of length 3 will result in an isosceles triangle with the largest area?
119. A cylindrical tin, with lid, is made from a thin sheet of metal of area S . The radius, height and volume of the tin are r , h and v respectively.
- Find S in term of v and r .
 - Given that, r – can vary and that there are is no wastage of metal, show that for a given v , area S is a minimum when $h = 2r$.
120. Find the dimension of the rectangle of largest area which can be inscribed in a circle whose radius is a .
121. Show that the volume of the largest cone, that can be inscribed inside a sphere of radius R is $\frac{32\pi}{81}R^3$.

122. A symmetric rain gutter cross-section is shown by the adjacent diagram. $AB = CD = 3$ units. Let θ ($0 < \theta < \frac{\pi}{2}$) be the inclination of the sides AB and CD to the horizontal.



Show that the area of the gutter $ABCD$ is $9 \sin \theta + \frac{9}{2} \sin 2\theta$. What angle of θ would make the cross-sectional area of $ABCD$ maximum?

123. A frame of a window is made by connecting a rectangular sector and semicircular sector as shown by the diagram. Radius of the semicircular sector is r and the height of the window is h . If the area of the window should be 500 square units, show that the perimeter p of the window is given by $p = \left(2 + \frac{\pi}{2}\right)r + \frac{500}{r}$ units. Determine the values of h and r so that the perimeter of the window is maximum.



PART - 4

Rate of Change.

- Summary of the Results.

Let $y = f(x)$ be a relation between two variables x and y . Then :

i. $\frac{dy}{dx}$ denotes the rate of change of y with respect to x .

ii. Rates of change of x and y with respect to the time t are given by the

relation: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$.

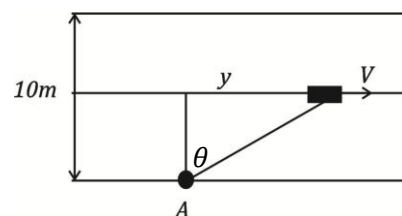
124. A particle moves along the curve $y = \frac{2}{3}x^3 + 1$. Find the points on the curve at which the y - co-ordinate is changing twice as fast as the x - co-ordinate.
125. A particle moves along a curve $6y = x^3 + 2$. Find the points on the curve at which the y - co-ordinate is changing 8 times as fast as the x - co-ordinate.
126. The radius of a circle is increasing at the rate of 7 cm/sec. Determine the rate at which the circumference of the circle is increasing.
127. The radius of a circle plate is increasing at the rate of 0.25 cm/sec. At what rate is the area increasing when the radius of the plate is 20 cm?
128. The side of a square is decreasing at the rate of 0.4 cm/sec. Find the rate of decrease of the,
- i. Area of the square
 - ii. Perimeter of the square, when the side of the square is 15 cm.
129. The radius of a spherical soap bubble is increasing at the rate of 0.05 cm/sec. Find the rate of increase in its volume when it radius is 4 cm.
130. The radius of a spherical balloon is decreasing at the rate of 0.2 cm/sec. Find the rate of decrease in its surface area when it radius is 25 cm.
131. A Balloon which remains Spherical, is being inflated by pumping in gas at the rate of 600 cm³/sec. Find the rate at which the radius of the balloon is increasing when the radius is 25 cm.

132. An edge of a variable cube is increasing at the rate of 0.24 cm/sec. How fast is the volume of the cube increasing when the edge is 15 cm long?
133. If the volume of an expanding cube is increasing at the rate of t^4 cu.cm/sec, then how fast is the surface area increasing, when the volume of the cube is 8 cm^3 ?
134. Find the length of the edge of the cube such that the rate of increase in its volume is the same as the rate of increase in its surface area.
135. Sand is being poured from a pipe at the rate of 8 cm/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-fourth of the radius of the base. How fast is the height of the sand cone increasing when the height is 6 cm?
136. Water is being pumped into a conical reservoir of height 15 meters and radius 6 meters, at the rate of 3 cubic meters per minute. How fast does the water level rise from a depth of 2.5 m?
137. Assume that a spherical rain drop evaporates at the rate proportional to its surface area. If its original radius is 3 mm and one hour later radius is 2 mm, find an expression for the radius of the rain drop at any time.
138. The area of a variable rectangle is increasing at the rate of $40 \text{ cm}^2/\text{sec}$. The length of the rectangle is always equal to the square of the breadth. At what rate is the length increasing at the instant when the breadth is 8 cm?
139. A ladder 5 m long is leaning against a wall, the foot of the ladder being 3 m away from the wall. If the lower end of the ladder is pulled away from the wall at the rate of 0.2 cm/sec, find the rate of decrease of the upper end along the wall.
140. A man 1.8 meters tall walks directly away from a lamp post, whose height is 9 meters, at a speed of 2 m/sec. Find the rate at which his shadow lengthens.
141. A particle moves along a straight line so that after t seconds its distance x meters from a fixed point O on the line is given by $x = t^3 - 3t^2 + t$. Find the velocity of the particle at $t = 3$.

142. A boy start walking north at a speed of 1.5 ms^{-1} , and a girl starts working west at the same point P at the same time at a speed of 2 ms^{-1} . At what rate is the distance between the boy and the girl increasing 6 seconds later?

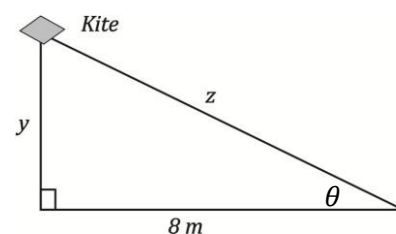
143. A ladder 15 feet long rests against a the vertical wall. Its top slides down the wall while its bottom moves away along the level ground at a speed of 2 feet per seconds. How fast is the angle between the top of the ladder and the wall changing when the angle is $\left(\frac{\pi}{3}\right)$ radians?

144. A person (A) situated at the edge of the river observes the passage of a speed boat going downstream. The boat travels exactly through the middle of the river. The river is 10 m wide. When the boat is at $\theta = 60^\circ$ (see figure) the observer measures the rate of change of the angle θ to be 2 radians per second. What is the speed, v of speed boat at the instant?



145. It kite is rising vertically at the constant speed of 2 ms^{-1} from a location at ground level which is 8 m away from the person handling the string of the kite.

- Let z be the distance from the kite of the person
- Find the rate of change of z with respect to time t when $z = 10$.
- Let θ be the angle the string makes with the horizontal. Find the rate of change of θ with respect to time t when the kite is $y = 6 \text{ m}$ above ground.



146. A spherical balloon is being inflected so that its volume is increasing at the constant rate of $20 \text{ cm}^3 \text{ s}^{-1}$. Show that the rate at which the radius increasing is $\left(\frac{5}{\pi r^2}\right) \text{ cms}^{-1}$, where r is the instantaneous radius of the balloon. Show also that, the rate at which the surface area increasing is $10 \text{ cm}^2 \text{ s}^{-1}$, when the radius of the balloon is 4 cm.

147. The height of rectangular box is increasing at a rate of 2 meters per second while the volume is decreasing at a rate of 5 cubic meters per seconds. If the base of the box is a square, at what rate is one of the sides of the base decreasing, at the moment when the base area is 64 square meters and the height is 8 meters?

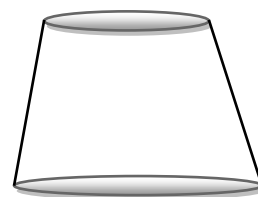
148. Water is poured in to a vessel in the form of a hollow cone of semi-vertical angle 45° , with vertex down words and axis vertical, so that the depth of water in the vessel increases at a uniform rate of 4 cms^{-1} . Find the rate at which the volume of water in the vessel increasing after two seconds.
149. A Conical tank whose semi vertical angle is $\left(\frac{\pi}{4}\right)$, with axis vertical and vertex downwards, allows water to fill to it through a small hole at its vertex, at the rate of $10 \text{ cm}^3 \text{ s}^{-1}$. At what rate is the
- Water level increasing?
 - the wetted surface increasing, when the depth of what is 5 cm.

Past Paper Questions

150. Let $f(x) \equiv \frac{1+2x}{x(x+1)}$.

Draw the rough graph of $y = f(x)$. By considering the intersection of the graph $y = f(x)$ and a straight-line $y = mx$, show that the third degree equation $mx^3 + mx^2 - 2x - 1 = 0$ has,

- 3 real roots when $m > 0$
 - Only one real root when $m \leq 0$
151. A straight tree trunk of circular cross section is 5m in length. This tree trunk is in the shape of a base of a cone. The radius of one base is 1.5 m and the other one's radius is 0.5 m. Show that the length of the cylindrical rod with the maximum volume that can be cut out from this tree trunk is 2.5 m.



152. Let $f(x) = \frac{x^2+x}{(x+2)^2}$; for $x \neq -2$.

Show that the increasing intervals and decreasing interval of $f(x)$.

Now, by using $f''(x) = \frac{-6x}{(x+2)^4}$, find the co-ordinates of the point of inflection of the graph of $y = f(x)$. By clearly indicating the asymptotes, turning point, point of inflection and intercepts on x and y axes, sketch the graph of $y = f(x)$.

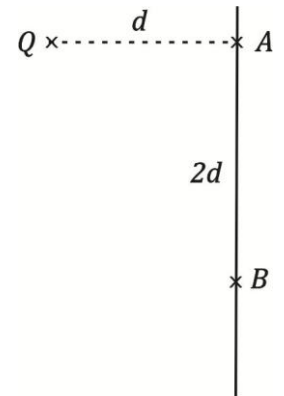
What is the vertical translation (displacement) that should be done, along the positive direction of any axis, for the above graph to be non-negative?

If $y = g(x)$ is the function for that corresponding graph,

Show that, $g(x) = \frac{9x^2 + 12x + 14}{8(x+2)^2}; x \neq -2$.

Write down the coordinates of the minimum point and the point of inflection for $y = g(x)$.

153. A man is at a point Q in the sea which is at a closest distance d km from a point A on a straight shore. His house is situated at a point B , which is at a distance $2d$ km from A . (see the figure)



The man can swim with a uniform velocity $u \text{ km h}^{-1}$ and he can walk with a uniform velocity $\sqrt{5} u \text{ km h}^{-1}$. Show that the man should swim to a point at a distance $\frac{3d}{2}$ km from B , within A and B , on the straight shore if he wants to reach his house as soon as possible. Find the minimum time he takes to reach his house.

154. From a solid sphere is cut a right circular cylinder whose axis passes through the centre of the sphere prove that the volume of the cylinder cannot exceed $\frac{1}{\sqrt{3}}$ times that of that sphere.
155. Three towns A , B and C are connected by two straight roads AB and BC such that $\hat{ABC} = \frac{\pi}{2}$, $AB = 15$ km and $BC = 50$ km. There is a proposed project to build another straight road connecting the town A to a place D on the road BC . Maximum speeds allowed for a car on the section DC is 50 km h^{-1} and on the proposed road AD is 40 km h^{-1} .

If D is located x km from the town A . Find $T(x)$ the total time taken in hours by a car to travel from A to C via D , assuming that it travels at the maximum permitted speed.

156. Given that $y = \frac{2x}{1+x^2}$, find the values of x which $\frac{dy}{dx} = 0$. Considering only the behavior of the first derivative, investigate the nature of those stationary values of y . Sketch the curve $y = \frac{2x}{1+x^2}$.

157. A rectangular poster is to be made so that it displays a rectangular printed region of area 972 cm^2 surrounded by a left and right margins of 6 cm each, and a top and bottom margins of 8 cm each.

Find the dimensions of the poster with the least area.

158. A rectangular box is to be made having a capacity of 256 cm^3 , with the square base, but without a lid. Find the dimensions of the cheapest box if the material for rectangular sides cost 8 times as much as centimeter as the material for the base.

159. Let $y = (1 + 4x^2) \tan^{-1}(2x)$.

Show that,

i. $(1 + 4x^2) \frac{dy}{dx} - 8xy = 2(1 + 4x^2)$ and ii. $(1 + 4x^2) \frac{d^2y}{dx^2} - 8y = 16x$

Find $\left(\frac{d^2y}{dx^2}\right)_{x=0}$

160. A closed right circular cylinder is to be made such that its volume is $1024 \pi \text{ cm}^3$. Find the radius of the cylinder that will make its total surface area a minimum.

161. Let $f(x) = 2x^3 + ax^2 + bx$ for $x \in \mathbb{R}$.

Where a and b are real constants. Suppose that $f'(3) = 12$ and $f''(3) = 18$, where f' and f'' have the usual meaning. Find the values of a and b .

For these values of a and b , sketch the graph of $y = f(x)$ indicateing the turning points.

Hence, find the number of solution of the equation $2x^2 + ax + b = \frac{3}{x}$.

162. Let $f(x) = \frac{x^2}{x^3-1}$ for $x \neq 1$.

Show that $f(x) = \frac{-x(x^3+2)}{(x^3-1)^2}$ for $x \neq 1$ and deduce that the graph of $y = f(x)$ has turning

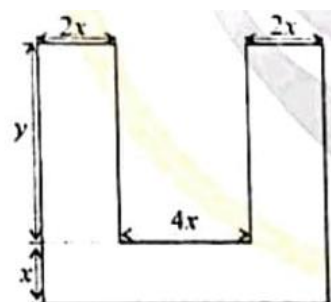
points at $(0, 0)$ and $\left(-2^{1/3}, \frac{4^{1/3}}{3}\right)$. Sketch the graph of $y = f(x)$ indicating the turning points and the asymptotes.

163. A garden whose boundary consist of eight straight line segments meeting at right angles is shown in the diagram. The dimensions in meters of the garden of indicate there. The area of the garden is given to be 800 m^2 . Expressly in terms of x and show that the perimeter P of the garden, measured in meters, is given by

$$P = \frac{800}{x} + 10x \text{ and this formula for the perimeter is valid only}$$

for $0 < x < 10$.

Hence find the minimum value of the perimeter of the garden.



164. By considering only the first derivative find the minimum and maximum values of $\frac{x^3}{x^4+27}$.

Sketch the graph of $y = \frac{x^3}{x^4+27}$.

Hence, find for what values of k , the equation $kx^4 - x^3 + 27k = 0$.

Where k is real, has

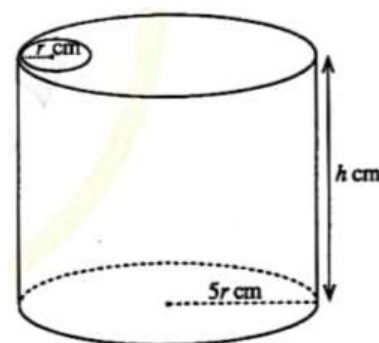
- Two coincident real roots,
 - Three coincident real roots,
 - two distinct real roots
 - no real roots.
165. Consider a rectangle $ABCD$ with $AB = a$ and $BC = b (< a)$. Let P be a movable point on CD . The length of $AP + PB$ is $L(x)$, Where $DP = x$.
- Show that $L(x) = \sqrt{x^2 + b^2} + \sqrt{(a-x)^2 + b^2}$.
- Find the minimum length of $L(x)$ and the position of P on CD corresponding to this minimum length. Also, find the maximum length of $L(x)$.

166. Let $f(x) = \frac{(x-3)^2}{x^2-1}$ for $x \neq \pm 1$.

Show that $f'(x)$, the derivative of $f(x)$ is given by $f'(x) = \frac{2(x-3)(3x-1)}{(x^2-1)^2}$. Write down the equations of the asymptote of $y = f(x)$.

Find the coordinates of the point at which the horizontal asymptote intersects the curve $y = f(x)$. Sketch the graph of $y = f(x)$ indicating the asymptotes and the turning points.

167. A thin metal container, in the shape of a right circular cylinder of radius $5r$ cm and height h cm has a circular lid of radius $5r$ cm with a circular hole of radius r cm. (See the figure.) The volume of the container is given to be 245π cm³. Show that the surface area of S cm² of the container with the lid containing the hole given by $S = 49\pi \left(r^2 + \frac{2}{r}\right)$ for $r < 5$. Find the value of r such that S is minimum.



168. Let $y = x \sin \frac{1}{x}$ for $x \neq 0$. Show that,

i. $x \frac{dy}{dx} - y = \cos \frac{1}{x}$ and ii. $x^4 \frac{d^2y}{dx^2} + y = 0$

169. Let $f(x) = \frac{2x^2+1}{(x-1)^2}$ for $x \neq 1$.

Find the first derivative and the turning point of $f(x)$. Sketch the graph of $y = f(x)$, indicating the turning point and the asymptotes.

170. In the given figure, $ABCD$ is a trapezium with parallel sides BC and AD . Length of its sides, measured in centimeters are given by, $AB = CD = a$, $BC = b$ and $AD < b < 2x$, where $0 < x < a$. BE and CF are the perpendicular drawn from the vertices B and C , respectively on to the side AD .

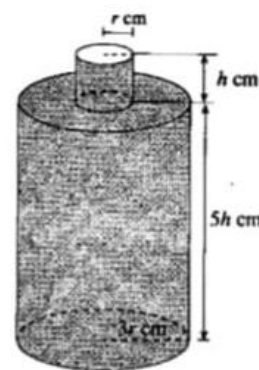
Show that the area $S(x)$ of the trapezium $ABCD$ is given by $S(x)$ in square centimeters. If a $\sqrt{6}$ and show further that $S(x)$ is maximum for a certain value of x , and find the value of x and the maximum area of the trapezium.

171. Let $f(x) = \frac{16(x-1)}{(x+1)^2(3x-1)}$ for $x \neq -1, \frac{1}{3}$.

Show that $f'(x)$, the derivative of $f(x)$, is given by $f'(x) = \frac{-32x(3x-5)}{(x+1)^3(3x-1)^2}$ for $x \neq -1, \frac{1}{3}$.

Sketch the graph of $y = f(x)$ indicating the asymptotes and the turning points. Using the graph, find the values of $k \in \mathbb{R}$ such that the equation $k(x+1)^2(3x-1) = 16(x-1)$ has exactly one root.

172. A bottle with the volume of $391\pi \text{ cm}^3$ is to be made by removing a disc of radius $r \text{ cm}$ from the top face of a closed hollow right circular cylinder of radius $3r \text{ cm}$ and height $5h \text{ cm}$, and fixing an open hollow right circular cylinder of radius $r \text{ cm}$ and height $h \text{ cm}$, as shown in the figure. It is given that the total surface area $S \text{ cm}^2$ of the bottle is $S = \pi r(32h + 17r)$. Find the value of r such that S is minimum.



173. Let $f(x) = \frac{x^2}{(x-1)(x-2)}$ for $x \neq 1, 2$.

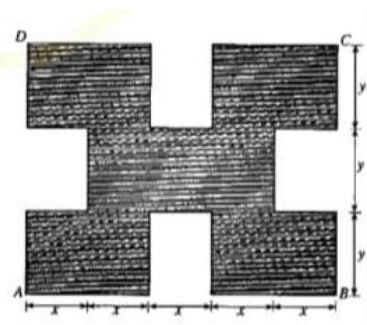
Show that $f'(x)$, the derivative of $f(x)$ is given by $f'(x) = \frac{x(4-3x)}{(x-1)^2(x-2)^2}$ for $x \neq 1, 2$. Sketch the graph of $y = f(x)$ indicating the asymptotes and the turning points. Using the graph, solve the inequality $\frac{x^2}{(x-1)(x-2)} \leq 0$.

174. The shaded region shown in the adjoining figure is of 385 m^2 .

This region is obtained by removing four identical rectangles each of length y meters and width x meters from a rectangle ABCD of length $5x$ meter and width $3y$ meters. Show that

$y = \frac{35}{x}$ and the perimeter P of the shaded region, measured in meters, is given by $P = 14x + \frac{350}{x}$ for $x > 0$.

Find the value of x such that P is minimum.



175. Let $f(x) = \frac{4x+1}{x(x-2)}$ for $x \neq 0, 2$.

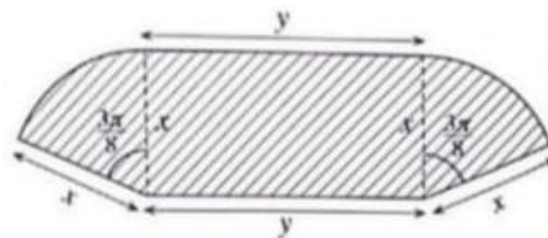
Show that $f'(x)$, the derivative of $f(x)$ is given by $f'(x) = -\frac{2(2x-1)(x+1)}{x^2(x-2)^2}$ for $x \neq 0, 2$.

Hence, find the intervals on which $f(x)$ is increasing and the intervals on which $f(x)$ is decreasing.

Sketch the graph of $y = f(x)$ indicating the asymptotes, x -intercept and the turning points.

Using this graph, find all real values of x satisfying the inequality $f(x) + |f(x)| > 0$.

176. The shaded region S of the adjoining figure shows a garden consisting of a rectangle and two sector of a circle each subtending an angle $\frac{3\pi}{8}$ at the centre. Its dimensions, in metres, and shown in the figure. The area of S is given to be 36 m^2 .



Show that the perimeter $p \text{ m}$ of S is given by $p = 2x + \frac{72}{x}$ for $x > 0$ and that p is minimum when $x = 6$.

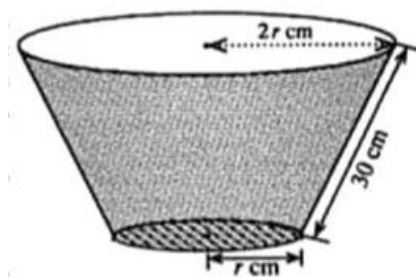
177. Let $f(x) = \frac{9(x^2-4x-1)}{(x-3)^3}$ for $x \neq 3$.

Show that $f'(x)$, the derivative of $f(x)$, is given by $f'(x) = -\frac{9(x+3)(x-5)}{(x-3)^4}$ for $x \neq 3$. Sketch

the graph of $y = f(x)$ indicating the asymptotes, y -intercept and the turning points. It is given that $f''(x) = \frac{18(x^2-33)}{(x-3)^5}$ for $x \neq 3$.

Find the x -coordinates of the points of inflection of the graph of $y = f(x)$.

178. The adjoining figure shows a basin in the form of a frustum of a right circular cone with a bottom. The slant length of the basin is 30 cm and radius of the upper circular edge is twice the radius of the bottom. Let the radius of the bottom be r cm.



Show that the volume $V \text{ cm}^3$ of the basin is given by $V = \frac{7}{3}\pi r^2 \sqrt{900 - r^2}$ for $0 < r < 30$. Find the value of r such that volume of the basin is maximum.

179. Let $f(x) = \frac{px+q}{(x-1)(x-2)}$ for $x \in \mathbb{R} - (1, 2)$, where $p, q \in \mathbb{R}$. It is given that the graph of $y = f(x)$ has a stationary point at $(0, 1)$. Show that $p = -3$ and $fq = 2$.
For these values of p and q , show that $f'(x)$, the derivative of $f(x)$, is given by $f'(x) = \frac{x(3x-4)}{(x-1)^2(x-2)^2}$ for $x \neq 1, 2$, and find the intervals on which $f(x)$ is decreasing and the intervals on which increasing.

Sketch the graph of $y = f(x)$ indicating the asymptotes and the turning points. Hence find the number of solutions to the equation $x^2(x-1)(x-2) = 2-3x$.

180. A cylinder with a top and a bottle is made to have a volume of $1024\pi \text{ cm}^3$. Let r cm be the radius of the cylinder. Show that the total surface area $S \text{ cm}^2$ of the cylinder is given by $S = 2\pi \left(\frac{1024}{r} + r^2 \right)$ for $r > 0$.
Show that S is minimum when $r = 8$.

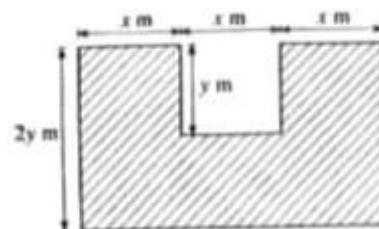
181. Let $f(x) = \frac{2x+3}{(x+2)^2}$ for $x \neq -2$.

Show that $f'(x)$, the derivative of $f(x)$, is given by $f'(x) = \frac{-2(x+1)}{(x+2)^3}$ for $x \neq -2$. Hence find the interval on which $f(x)$ increasing and the intervals on which $f(x)$ is decreasing. Also, find the coordinates of the turning point of $f(x)$.

It is given that $f'(x) = \frac{2(2x+1)}{(x+2)^4}$ for $x \neq -2$.

Find the coordinates of the point of inflection of the graph of $y = f(x)$. Sketch the graph of $y = f(x)$ indicating the asymptotes, the turning point and the point of inflection. State the smallest value of which $f(x)$ is one-one on $(k, \dots)$

182. The shaded region shown in the figure is of area 45 m^2 . It is obtained by removing a rectangle of length $x \text{ m}$ and width $y \text{ m}$ from a rectangle of length $3x \text{ m}$ and width $2y \text{ m}$. Show that the perimeter $L \text{ m}$ of the shaded region is given by $L = 6x + \frac{54}{x}$ for $x > 0$.



Find the value of x such that L is minimum.

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